Jenkins Deployment Using Docker on Ubuntu Server

Overview

This project demonstrates how to deploy **Jenkins** inside a **Docker** container on an Ubuntu server. The setup uses SSH for remote access and exposes Jenkins via a web interface.

Environment

- OS: Ubuntu Server
- Server IP: 192.168.1.124
- Access Method: SSH (Git Bash on Windows)

Technologies

- **Jenkins**
- **Docker**
- **Linux**

Prerequisites

- Ubuntu server with network access
- SSH access to the server
- Sudo privileges

Step: Running Jenkins Container

The following command was used to start the **Jenkins** container using **Docker**, with port mapping and a persistent volume:

```
docker run -p 8080:8080 \

-p 50000:50000 \

-d -v jenkins_home:/var/jenkins_home \

jenkins/jenkins:lts
```

- -p 8080:8080 → Exposes Jenkins web UI
- -p 50000:50000 → Enables agent communication
- -v jenkins_home:/var/jenkins_home → Persists Jenkins data

- -d → Runs container in detached mode

```
ali-abdul-aleem@ali-abdul-aleem-OptiPlex-3020: ~
ali-abdul-aleem@ali-abdul-aleem-OptiPlex-3020:~$ docker run -p 8080:8080 \
> -p 50000:50000 -d -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
53c88f1dfeb7: Pull complete
4d2125ccf71c: Pull complete
69503609dc01: Pull complete
6da7907f70f4: Pull complete
6327a7366a6f: Pull complete
01938f7c040b: Pull complete
b7b154fd69a1: Pull complete
876217fdee20: Pull complete
fbae626c1b7a: Pull complete
b19825c62ed8: Pull complete
ec7478903548: Pull complete
e682a95ad036: Pull complete
Digest: sha256:c4098086090ca98491d4bf66182f5e3b015a8232f2acf2df209a212a5801aa8e
Status: Downloaded newer image for jenkins/jenkins:lts
85ebcb80d305b1304dd7405091f1d6420e65d8774202d8f93c2c0b5012214c9b
ali-abdul-aleem@ali-abdul-aleem-OptiPlex-3020:~$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED
   STATUS              PORTS                                       NAMES
85ebcb80d305   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   About a minute ago
   Up About a minute   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:5000
0->50000/tcp, [::]:50000->50000/tcp   cool_rubin
ali-abdul-aleem@ali-abdul-aleem-OptiPlex-3020:~$ docker exec -it 85ebcb80d305 bash
jenkins@85ebcb80d305:/$ cat /var/jenkins_home/secrets/initialAdminPassword
565c0ff77705495fb7716bba93c31095
jenkins@85ebcb80d305:/$ |
```

Step: Verifying Container and Retrieving Admin Password

The container was successfully started and verified using:

docker ps

To retrieve the initial admin password required for setup:

docker exec -it <container_id> bash

cat /var/jenkins_home/secrets/initialAdminPassword

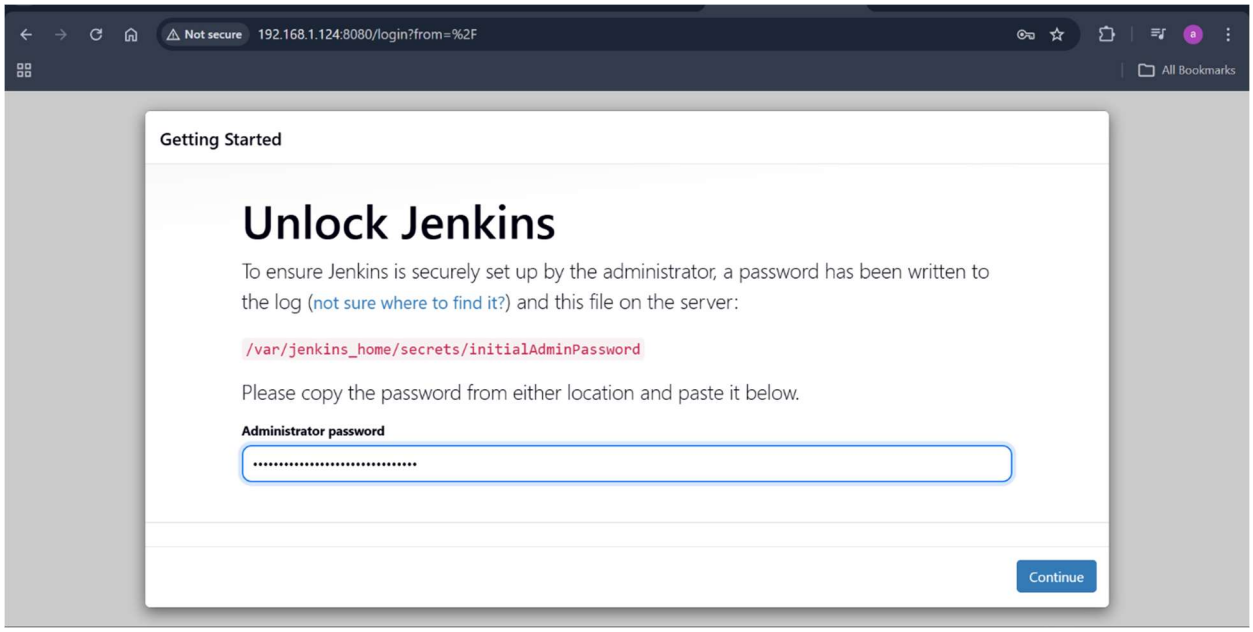
This password is required to unlock Jenkins on first access.

Step: Unlock Jenkins (Web Interface)

After starting the container, Jenkins is accessible via http://192.168.1.124:8080

The initial setup screen prompts for an administrator password.

- The password is located at: /var/jenkins_home/secrets/initialAdminPassword
- Paste the retrieved password into the input field to continue setup

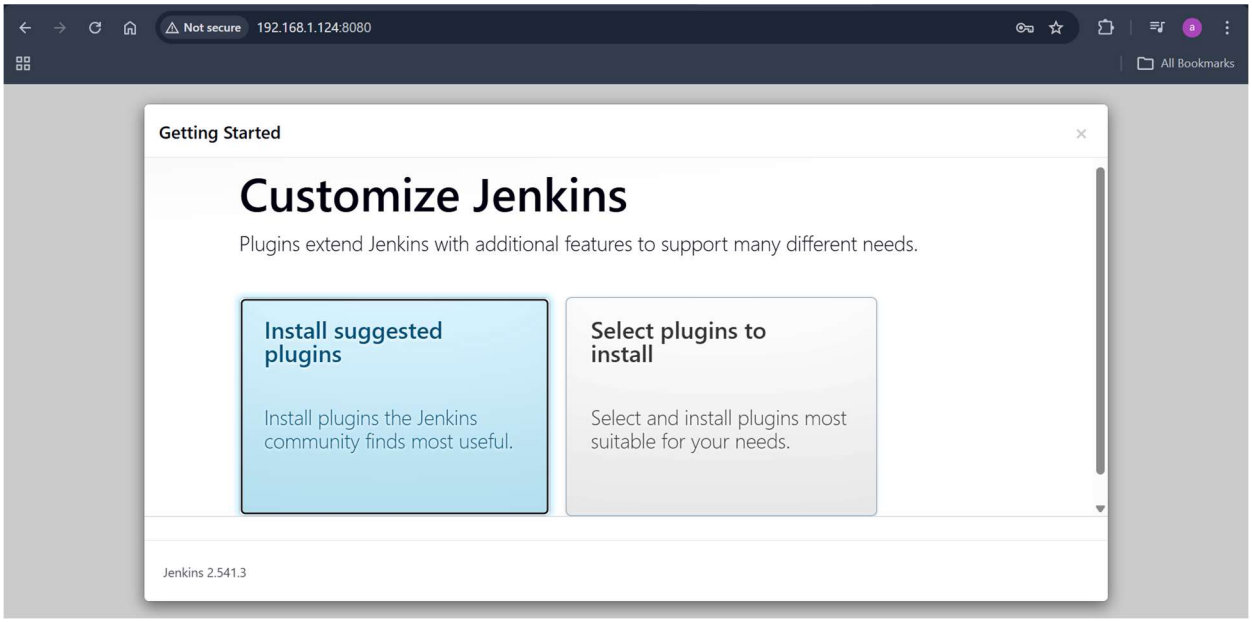


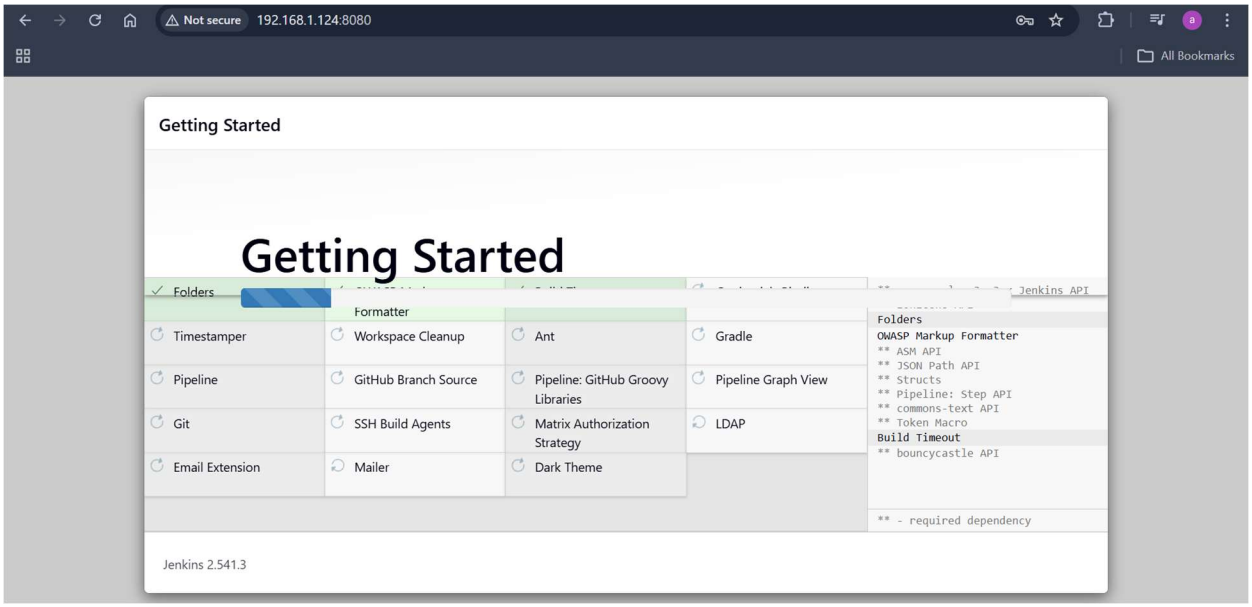
Step: Customize Jenkins

Once unlocked, Jenkins provides plugin installation options:

- **Install suggested plugins** → Recommended default setup
- **Select plugins to install** → Manual selection

For this setup, the **recommended plugins** option can be selected to quickly initialize the environment with commonly used tools.





Notes

- A Docker volume (jenkins_home) is used to ensure data persistence
- The container is accessible on port 8080 from the host machine
- Initial setup only needs to be completed once